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Photo-enhanced spillover hydrogenation over semiconductor-supported Pd nanocatalysts

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Spillover hydrogenation—a process by which reduction occurs via spilled hydrogen species—is an emerging strategy for selective hydrogenation reactions. Methods to promote spillover hydrogenation on supported Pd nanocatalysts, such as encapsulating Pd nanoparticles or using porous channels, often reduce the availability of Pd sites or increase mass transfer resistance. Here we show that light irradiation can directly induce site separation to enable spillover hydrogenation over semiconductor-supported Pd nanocatalysts, without impairing catalytic activity. Photogenerated electrons accumulated on Pd nanoparticles suppress unsaturated substrate adsorption, ensuring that the Pd exclusively facilitates H_2 dissociation. Photogenerated holes left on the support enhance substrate adsorption and active hydrogen migration. Here, this photo-enhanced spillover hydrogenation is validated across multiple catalytic systems (Pd/$SrTiO_3$, Pd/ZnO, Pd/PCN and Ru/MnO_x), demonstrating that this effect is not material-specific, but is a generalizable interfacial phenomenon. These findings suggest that light irradiation is a widely applicable methodology for improving spillover hydrogenation efficiency.

Hydrogenation reactions are industrially important for the production of petrochemicals, agrochemicals and fine chemicals[1–3]. Supported Pd nanocatalysts have been used extensively to catalyse these reactions because of the unique $4d^{10}$ valence electron configuration of Pd, which facilitates the activation and dissociation of H_2 into active hydrogen (H*) species[4,5]. However, chemoselective hydrogenations over supported Pd nanocatalysts remain challenging, especially when two or more unsaturated substrates are present in the reaction system[6–8]. For example, a serious overhydrogenation of alkenes is likely to occur during the semihydrogenation of alkynes on these nanocatalysts[9], because unsaturated substrates such as alkenes and alkynes can strongly adsorb on Pd nanoparticles along with the H_2, due to the metal's ability to form strong interactions with π electrons[10,11]. Consequently, once H* species are generated, they are immediately captured by nearby adsorbed unsaturated substrates[12,13]. The occurrence of H_2 dissociation and substrate hydrogenation on the same Pd nanoparticle makes the kinetic control of chemoselective hydrogenation over supported Pd nanocatalysts rather difficult[14–16].

Hydrogen spillover is a phenomenon observed in catalysis where hydrogen atoms migrate from metals to supports[17–23]. Recently,

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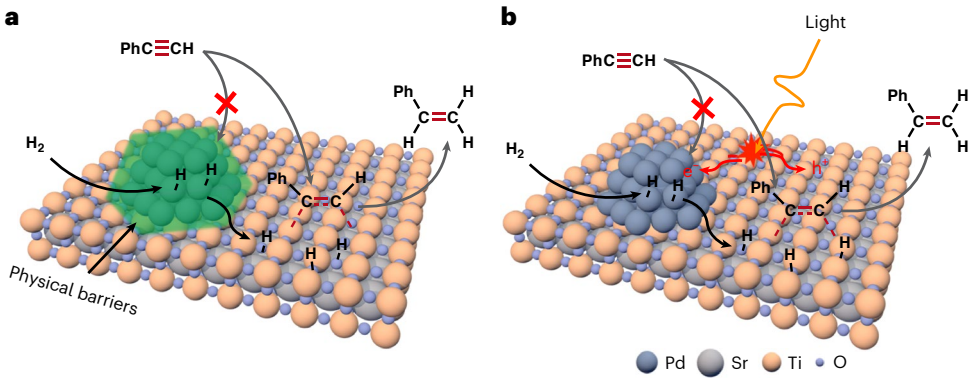

Fig. 1 | Illustration of spillover hydrogenation. a,**b**, Traditional (**a**) and photo-enhanced (**b**) strategies.

approaches for the hydrogenation of a substrate utilizing hydrogen atoms spilled over from the metal onto the support have been developed to improve selectivity, in what is referred to as 'spillover hydrogenation'[24–28]. To ensure that the H_2 and substrates are selectively adsorbed on the Pd nanoparticles and the support surface, respectively, Pd nanoparticles have been either encapsulated with shells[29–31], embedded in porous channels[32] or decorated with additive molecules[33] to reduce or completely suppress substrate adsorption on their surfaces (Fig. 1a). These strategies lead to a substantial improvement in selectivity towards the target products. However, the inevitable shielding of Pd active sites and/or increased mass transfer resistance greatly limit catalytic performance for H_2 dissociation and H* generation, leading to a noticeable decrease in catalytic activity for selective hydrogenation reactions using these approaches. To address this issue, it is crucial to develop strategies that ensure the complete exposure of supported Pd nanoparticles, allowing them to be fully accessible to H_2 to catalyse effective spillover hydrogenation.

In this Article, we show that light irradiation can directly induce site separation of the H_2 activation (on bare Pd nanoparticles) and hydrogenation reactions (on the semiconductor support) by using bandgap excitation to tailor the electronic structure and coordination environment of the bare Pd active sites (Fig. 1b)[8]. Using light irradiation with energies greater than the bandgap of the semiconductor, photogenerated electrons transfer rapidly to the supported Pd nanoparticles and accumulate on their surfaces. This electron-rich surface structure can effectively inhibit the adsorption of unsaturated substrates such as alkynes[11,34], and thus allows the Pd nanoparticles to exclusively adsorb and activate H_2 molecules. Moreover, the increase in electron density on the surface Pd nanoparticles facilitates dissociation of the H_2 into H* species[35], which in turn promotes efficient hydrogen spillover from metals to the support[36]. Meanwhile, the photogenerated holes left on the support can enhance substrate adsorption while also weakening the H*–support interaction to promote the migration of H* species[37]. As a result, spatial separation of H_2 activation sites on the bare Pd nanoparticles and unsaturated substrate hydrogenation sites on the semiconductor support is achieved by simply introducing light irradiation into the reaction system. This is rather different from the phenomenon recently observed in, for example, Pd-decorated Al nanocrystals, where H_2 activation and substrate hydrogenation took place on the same Pd nanoparticles, and the light irradiation enhanced H_2 desorption to improve selectivity[38]. Experimentally, this photo-enhanced spillover hydrogenation process has been evaluated using Pd nanoparticles supported on strontium titanate ($Pd/SrTiO_3$) as a model catalyst, showing good selectivity during the semihydrogenation of alkynes to alkenes, while maintaining high catalytic activity. The effect is also observed in other catalytic systems, including Pd nanoparticles supported on ZnO and carbon nitride semiconductors, as well as Ru nanoparticles supported on MnO_x. We anticipate that this phenomenon may contribute broadly to improving the efficiency of spillover hydrogenation strategies.

Results

Photo-enhanced selectivity during semihydrogenation

Supported Pd nanocatalysts were prepared by means of a photodeposition method[39,40]. Figure 2a,b and Supplementary Fig. 1 present aberration-corrected high-angle angular dark-field scanning transmission electron microscopy (AC-HAADF-STEM) images of $Pd/SrTiO_3$. Hemispherical Pd nanoparticles with dimensions of ~5 nm were homogeneously distributed on the $SrTiO_3$ surface, and the lattice fringes with interplanar spacings of 0.23 and 0.19 nm were assigned to the (111) and (002) planes of metallic Pd. In X-ray diffraction patterns (Supplementary Fig. 2), no diffraction peak originating from Pd species was found, further indicating that the Pd nanoparticles had been successfully deposited as ultrafine nanoparticles on the $SrTiO_3$ with uniform dispersion. The catalytic performance of $Pd/SrTiO_3$ was evaluated by the semihydrogenation of phenylacetylene (PA) to styrene using H_2. Figure 2c demonstrates the serious overhydrogenation of styrene to ethylbenzene on $Pd/SrTiO_3$ during PA hydrogenation in the dark, showing a typical catalytic behaviour of supported Pd nanoparticles in selective hydrogenation reactions[15]. Interestingly, when light irradiation was simply introduced to this hydrogenation reaction system, the overhydrogenation was effectively suppressed while maintaining high catalytic activity. This photo-enhanced selectivity was further demonstrated by recycling experiments with $Pd/SrTiO_3$ in a hydrogenation reaction (Supplementary Fig. 3), as well as with light on/off tests for assessing catalytic performance (Supplementary Fig. 4). Supplementary Fig. 5 demonstrates the photo-induced inhibition of further hydrogenation of styrene by conducting the reaction in an environment with only styrene as the unsaturated substrate. The result shows that the hydrogenation of styrene to ethylbenzene is substantially reduced under light irradiation. This observation clearly demonstrates that the light irradiation indeed dramatically suppresses the overhydrogenation process, thus exhibiting a unique photo-enhanced selectivity towards PA semihydrogenation. Supplementary Fig. 6 shows that this photo-enhanced selectivity is also observed on $SrTiO_3$-supported Pd active phases ranging from single atoms to nanoparticles of various sizes, suggesting that the photo-induced enhancement in selectivity is consistent across different Pd structures. Supplementary Fig. 7 presents the catalytic behaviour of PA semihydrogenation while irradiating the reaction system with light of different wavelengths. As summarized in Fig. 2d, the trend of selectivity towards styrene matches well with the optical absorption of $SrTiO_3$, exhibiting a typical wavelength-dependent selectivity. These results indicate that the selective hydrogenation of PA is closely dependent on the photoexcitation of the $SrTiO_3$ semiconductor, and photogenerated carriers from $SrTiO_3$ are involved and contribute to the semihydrogenation reaction.

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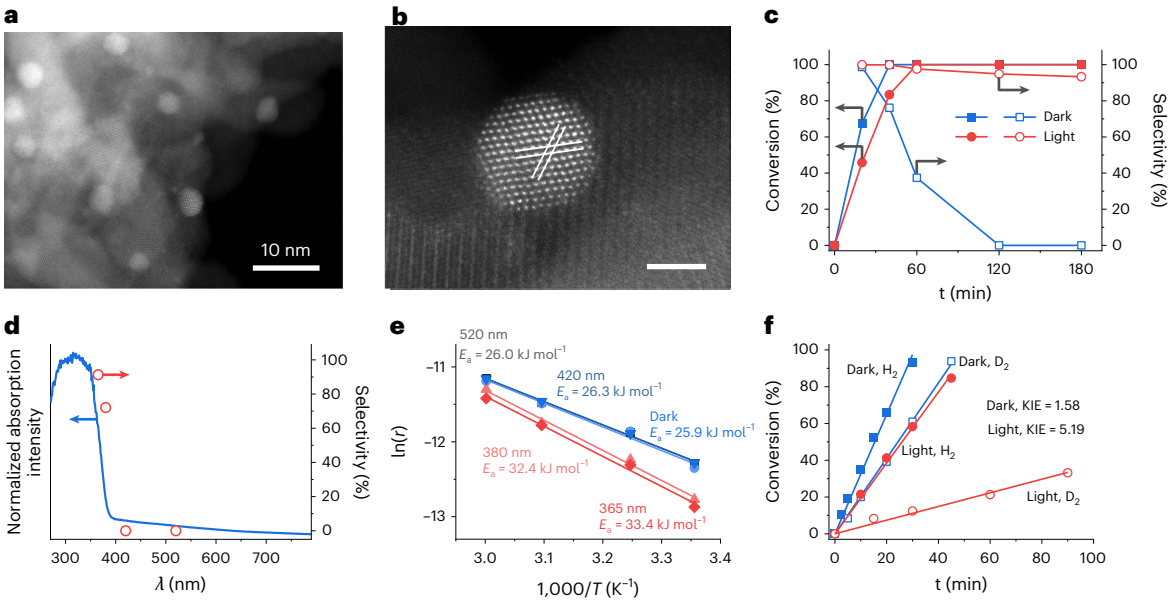

Fig. 2 | Structural characterization and catalytic performance of Pd/SrTiO$_3$. **a**,**b**, AC-HAADF-STEM images of Pd/SrTiO$_3$. **c**, Time course of PA conversion and styrene selectivity over Pd/SrTiO$_3$, with or without light irradiation. **d**, Action spectrum of wavelength-dependent styrene selectivity over Pd/SrTiO$_3$. **e**, Arrhenius plots for PA hydrogenation over Pd/SrTiO$_3$ with light of different wavelengths. **f**, KIE values of PA hydrogenation over Pd/SrTiO$_3$.

To rule out the influence of photo-thermal contributions, the reaction temperature was maintained around 25 °C using a water bath equipped with circulating water (Supplementary Fig. 8). The involvement of photogenerated electron–hole pairs in the hydrogenation reaction was further confirmed by control experiments using insulator-supported Pd nanocatalysts, such as Pd/Al$_2$O$_3$. Supplementary Fig. 9 shows that there is no evident difference in catalytic performance for Pd/Al$_2$O$_3$ with or without light irradiation, suggesting that the unique photo-enhanced selectivity closely depends on semiconductor bandgap excitation.

A kinetics study of PA semihydrogenation was carried out to gain insights into this wavelength-dependent photo-enhanced selectivity. Figure 2e shows that the apparent activation energy (E_a) for PA semihydrogenation over Pd/SrTiO$_3$ differs based on whether the support undergoes bandgap excitation by the incident light. Specifically, when SrTiO$_3$ is photoexcited by light with energies greater than its bandgap of 3.2 eV (for example, wavelengths of 365 and 380 nm; Supplementary Fig. 10), the E_a is ~33 kJ mol^{-1}. In contrast, when exposed to light irradiation with energies lower than the bandgap (for example, wavelengths of 420 and 520 nm), the E_a is ~26 kJ mol^{-1}, similar to the value obtained in the dark. These results indicate that light-induced bandgap excitation of the support substantially affects the energy barrier for PA semihydrogenation over Pd/SrTiO$_3$. Supplementary Fig. 11 presents the results of deuterium labelling experiments with deuterium gas (D_2), carried out to study the kinetic isotope effect (KIE) of PA semihydrogenation over Pd/SrTiO$_3$. The reaction in the dark was slowed by a factor of 1.58 due to the zero-point energy difference between isotopic isomers (Fig. 2f), consistent with the KIE values determined for PA hydrogenation on the surface of Pd nanoparticles[41]. In contrast, a larger KIE of 5.19 was obtained under light irradiation, suggesting that the hydrogenation mechanism for PA semihydrogenation over Pd/SrTiO$_3$ under light irradiation is rather different from that in the dark. The hydrogenation pathway was further studied using in situ Fourier-transform infrared (FTIR) spectra (Supplementary Fig. 12). Under both dark and light conditions, the characteristic peaks assigned to PA molecules adsorbed on Pd/SrTiO$_3$ disappeared quickly after injecting the H_2 molecules. This finding ruled out the possibility that the hydrogenation pathway had shifted from the Langmuir–Hinshelwood (LH) mechanism to the Eley–Rideal (ER) mechanism. Considering that the adsorption and dissociation of H_2 to produce H* was rapid on the well-exposed Pd nanoparticle surface, the larger KIE along with a higher E_a should suggest that hydrogen spillover and hydrogenation processes on SrTiO$_3$ may be involved in PA semihydrogenation under light irradiation[25,42]. To verify the contribution of SrTiO$_3$, pure SrTiO$_3$ was physically mixed with Pd/SrTiO$_3$. It was found that the activity of Pd/SrTiO$_3$ was readily enhanced by ~30% after mixing with 0.5 equiv. of SrTiO$_3$ under light irradiation, suggesting that PA hydrogenation had occurred on the SrTiO$_3$ (Supplementary Fig. 13).

Photo-induced strong metal–support interaction

Figure 3a and Supplementary Fig. 14 present Pd K-edge extended X-ray adsorption fine structure (EXAFS) spectra of Pd/SrTiO$_3$, with and without light irradiation. For Pd/SrTiO$_3$ in the dark, peaks at 1.5 and 2.6 Å are attributed to the contributions of Pd–O and Pd–Pd coordination, respectively. The Pd–O coordination results from oxidation of the Pd nanoparticles exposed to air. Under light irradiation, a photo-induced transformation from Pd–O to the metallic Pd–Pd bond was observed, with Pd–O coordination decreasing while the metallic Pd–Pd coordination increased. To rule out the possibility of particle growth under light irradiation, transmission electron microscopy (TEM) images of Pd/SrTiO$_3$ were collected at different reaction times. In Supplementary Fig. 15, no evident change in the size of the Pd nanoparticles is observed, providing solid evidence that no particle growth has occurred under light irradiation. The light-induced increase in metallic Pd–Pd coordination was further confirmed by X-ray absorption near-edge spectroscopy (XANES) of the Pd K-edge (Fig. 3b). Under light conditions, an evident shift of the absorption edge of the Pd K-edge to a lower-energy position is observed, indicating the electron-rich state of the Pd nanoparticles. These results suggest that electrons have accumulated on the Pd surface under light irradiation.

Figure 3c presents in situ X-ray photoelectron spectroscopy (XPS) Pd 3*d* results. Under dark conditions, the peak with a binding energy of 335.0 eV originates from Pd^0, and the peak at 336.6 eV was assigned to the Pd^{2+} species, indicative of the partial oxidation of Pd in air. Under light irradiation, a substantial rightshift of these binding energies is observed. This shift indicates that photogenerated electrons have been

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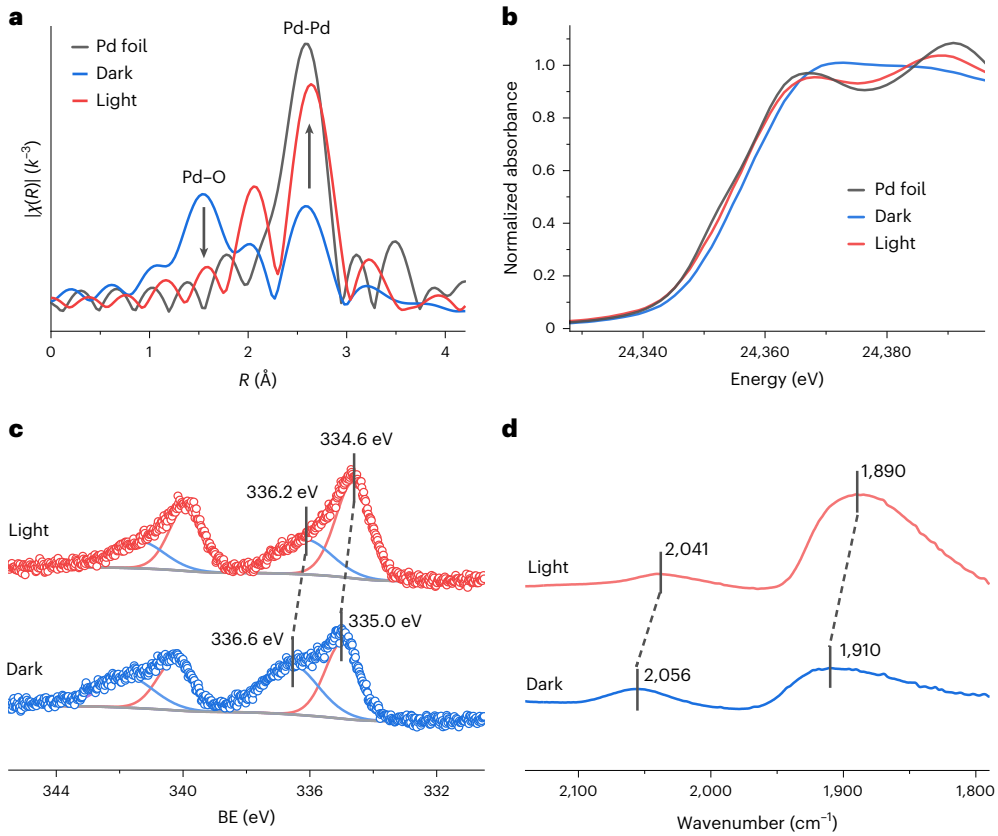

Fig. 3 | Photo-induced electrons accumulate on the Pd nanoparticles. a,**b**, Pd K-edge EXAFS spectra (**a**) and Pd K-edge normalized XANES spectra (**b**) of Pd/$SrTiO_3$. **c**, In situ Pd 3*d* XPS spectra of Pd/$SrTiO_3$ with and without light irradiation. BE, binding energy. **d**, CO-DRIFTS results for Pd/$SrTiO_3$ after light irradiation or the catalyst in the dark condition.

transferred rapidly from $SrTiO_3$ to the supported Pd nanoparticles, leading to increased electron density on the Pd nanoparticles. Figure 3d further confirms a photo-induced increase in electron density on the Pd surface (by in situ diffuse reflectance infrared Fourier-transform spectroscopy using CO as the probe molecule; CO-DRIFTS), with a redshift and enhancement of the CO adsorption peaks on Pd/$SrTiO_3$ observed under light irradiation[43]. In addition, slight positive shifts in the binding energies of the Sr 3*d* and Ti 2*p* XPS spectra are observed under light irradiation, suggesting that $SrTiO_3$ has become positively charged due to the retention of photogenerated holes on the support (Supplementary Fig. 16). This photo-induced charge redistribution, particularly the enrichment of electrons on the Pd nanoparticles, should facilitate spillover hydrogenation over Pd/$SrTiO_3$ (refs. 44,45), because the increased electron density on the Pd surface could suppress alkynes adsorption[46,47], allowing the bare Pd nanoparticles to be exclusively available for H_2 activation and H* generation.

Photo-induced selective adsorption of PA on $SrTiO_3$

The adsorption energies of PA on Pd nanoparticles and $SrTiO_3$ were calculated to be −1.58 and −0.46 eV, respectively, in the dark (Fig. 4a). Under light irradiation, as photogenerated electrons accumulate on the Pd to increase the surface electron density, the adsorption energy of PA on the Pd nanoparticles gradually decreases (Supplementary Fig. 17), transitioning from adsorption to non-adsorption (1.00 eV). Meanwhile, the adsorption energy of PA on $SrTiO_3$ increases to −0.71 eV due to the holes left on the $SrTiO_3$. This finding indicates that light irradiation evidently reduces PA adsorption on Pd nanoparticles while enhancing it on $SrTiO_3$, effectively shifting PA adsorption sites from the Pd nanoparticles to $SrTiO_3$ (Fig. 4b). This photo-induced site separation should be crucial for enabling spillover hydrogenation over Pd/$SrTiO_3$.

The adsorption of PA on Pd/$SrTiO_3$ was experimentally studied using in situ FTIR spectra (Fig. 4c). In the dark, the adsorption of PA on the catalyst surface was confirmed by the emergence of strong characteristic peaks assigned to PA (Supplementary Fig. 18), such as the peaks at 2,110 cm^{-1} for the stretching vibration of the C≡C bond and below 2,000 cm^{-1} for the out-of-plane bending vibration of the Ar–H bond in PA (line ii). Quinoline was used as a poison agent to selectively block the Pd nanoparticles (Pd/$SrTiO_3$–quinoline), as it remains intact during the PA semihydrogenation reaction (Supplementary Fig. 19). In Fig. 4d, these characteristic peaks are seen to become less intense (line iii), indicating that the adsorption of PA on Pd has been effectively suppressed due to the spatial restriction caused by the adsorbed quinoline[48]. This blocking of PA adsorption by quinoline was further confirmed by the rapid drop in catalytic activity observed for Pd/$SrTiO_3$–quinoline under dark conditions. Under light conditions, no evident change in intensity was observed for the characteristic peak at 2,110 cm^{-1} before and after the quinoline modification (line iv, Fig. 4c versus line vi, Fig. 4d). This suggests that the PA substrate is predominantly adsorbed on the $SrTiO_3$ rather than on the Pd nanoparticles under light irradiation, regardless of the presence of quinoline decoration on the Pd surface. In Supplementary Fig. 20, the photo-induced selective adsorption of PA on $SrTiO_3$ is further confirmed in temperature-programmed desorption profiles. Under light irradiation, the PA desorption peak is notably reduced on Pd/$SrTiO_3$, but increases on Pd/$SrTiO_3$–quinoline. The decreased PA adsorption observed in line iv of Fig. 4c compared to line ii indicates that light irradiation has effectively suppressed PA adsorption on the Pd nanoparticles, probably due to electrons accumulating on the Pd surface. Conversely, the enhanced PA adsorption seen in line vi compared to line iii in Fig. 4d suggests that light irradiation has facilitated PA adsorption on $SrTiO_3$, due to the holes left on the support.

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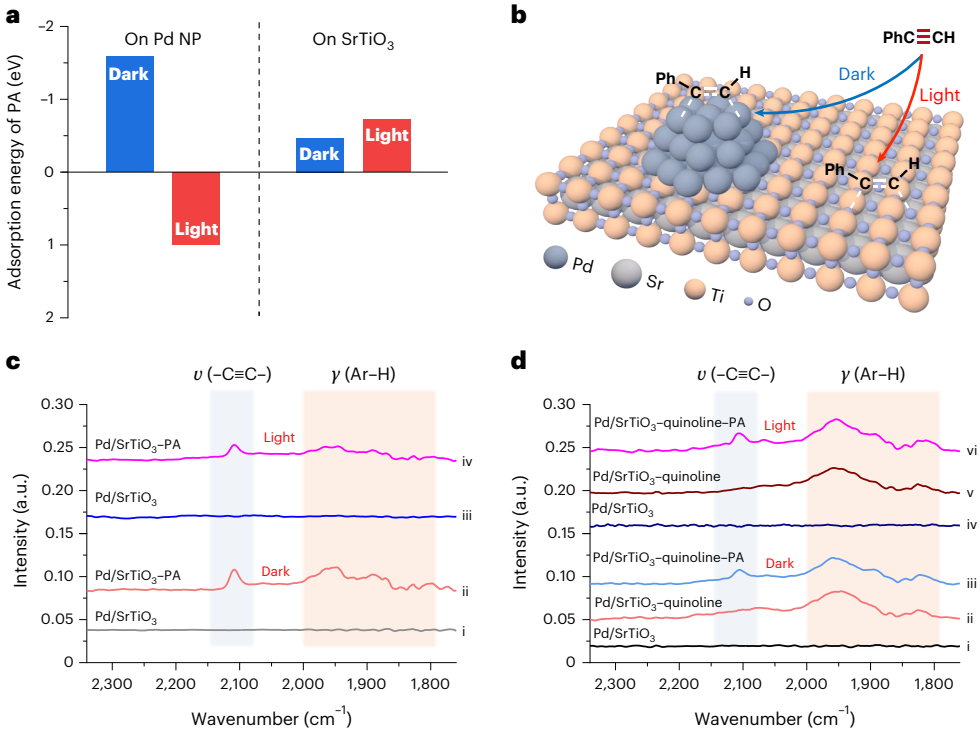

Fig. 4 | Adsorption behaviour of PA on Pd/$SrTiO_3$. a, Adsorption energy of PA on Pd nanoparticles and $SrTiO_3$, with or without light irradiation. **b**, Schematic diagram of PA adsorption on Pd nanoparticles and $SrTiO_3$. **c**,**d**, In situ PA-FTIR spectra of Pd/$SrTiO_3$ (**c**) and Pd/$SrTiO_3$–quinoline (**d**) with or without light irradiation. In **c**, the light blue band corresponds to the peak attributed to the stretching vibration of the C≡C bond in PA, while the light pink band represents the peaks assigned to the out-of-plane bending vibration of the Ar-H bond in PA.

These observations further confirm that light irradiation has reduced PA adsorption on the Pd nanoparticles while enhancing it on $SrTiO_3$, consistent with the conclusions derived from theoretical calculations (Fig. 4a). This photo-induced selective adsorption of PA on $SrTiO_3$ should be beneficial for the spillover hydrogenation of PA to styrene.

Photo-enhanced hydrogen spillover from Pd to $SrTiO_3$

WO_3 was next used as a colour indicator to study hydrogen spillover on Pd/$SrTiO_3$, as it is readily reduced by spilled H* species to form H_xWO_3 (Fig. 5a). As a result of intervalence charge-transfer transitions between W^{5+} and W^{6+} centres[49], the H_xWO_3 species displayed a strong blue colour. Under light irradiation, the physical mixture of Pd/$SrTiO_3$ and WO_3 (Pd/$SrTiO_3$–WO_3) changed rapidly to dark blue in an H_2 flow, indicating the migration of H* species from Pd/$SrTiO_3$ to WO_3 (ref. 49). Accordingly, a characteristic tailing absorption caused by H_xWO_3 species was clearly observed in UV–vis diffuse reflectance spectra (UV–vis DRS) for Pd/$SrTiO_3$–WO_3 after light irradiation (Fig. 5b). In contrast, a hydrogen spillover-induced colour change was not obvious for the Pd/$SrTiO_3$–WO_3 mixture in an H_2 flow without light irradiation. The observed slight colour change was attributed to the reduction of WO_3 by the H_2 gas rather than by the spilled H* species (Supplementary Fig. 21). These observations suggest that light irradiation efficiently promotes hydrogen spillover on Pd/$SrTiO_3$. This photo-enhanced hydrogen spillover was further studied by density functional theory (DFT) calculations. Supplementary Fig. 22 shows that H* could easily spill over to the $SrTiO_3$ with barriers of 0.01–0.46 eV, suggesting that hydrogen spillover from Pd nanoparticles to the $SrTiO_3$ surface is facilitated under light irradiation. This was further confirmed by electron paramagnetic resonance (EPR) spectra. In Fig. 5c, a distinct symmetrical peak at $g = 2.002$ is observed for Pd/$SrTiO_3$ after light irradiation in a H_2 flow; this is attributed to unpaired electrons trapped by oxygen vacancies[45]. These oxygen vacancies are created by the reduction of $SrTiO_3$ as a result of the spilled H* species.

The migration of H* species from Pd nanoparticles to $SrTiO_3$ was directly observed in deuterium-labelled in situ FTIR spectra (Fig. 5d), which allowed the movement of H* species on the Pd/$SrTiO_3$ nanocatalyst to be tracked. Under light irradiation, a strong peak assigned to the stretching vibration of the O–D bond emerged in the range of ~2,000–2,800 cm^{-1} following the injection of D_2 gas (Fig. 5d, line ii), directly indicating the occurrence of hydrogen spillover from Pd to $SrTiO_3$ (Fig. 5e). In contrast, this peak was absent when D_2 was injected in the dark (line iii). This finding suggests that light irradiation can effectively promote hydrogen spillover over Pd/$SrTiO_3$, owing to the electrons accumulating on Pd to facilitate H_2 dissociation and the holes being left on the $SrTiO_3$ to accelerate H*-species migration. This is rather different to previous observations on Pd-decorated Al nanocrystals, in which light irradiation mediated H_2 desorption[38] (Supplementary Fig. 23). The reactivity of spilled H* species on the $SrTiO_3$ surface towards PA hydrogenation was studied by injecting PA. In Fig. 5d, the characteristic peaks of PA are clearly seen for the sample pretreated with D_2 in the dark (line iv), indicating strong PA adsorption on Pd/$SrTiO_3$. For the sample pretreated with D_2 under light irradiation, the strong O–D vibration peak disappeared upon the injection of PA, and the peaks assigned to PA were difficult to detect (line v). This suggests that the consumption of O–D species should be attributed to the hydrogenation of PA on $SrTiO_3$. In Supplementary Fig. 24, the produced deuterated styrene can clearly be detected in the gas chromatography mass spectrometry, demonstrating that the spilled H* species on $SrTiO_3$ were indeed active for PA hydrogenation (Fig. 5e)[50]. In addition, no peak for styrene or ethylbenzene were detected in line v, suggesting that the styrene has easily desorbed from the $SrTiO_3$, as supported by the styrene FTIR spectra in Supplementary Fig. 25. The facile desorption of the styrene produced from the $SrTiO_3$ surface helps prevent further hydrogenation of styrene, thereby ensuring high selectivity for PA semihydrogenation.

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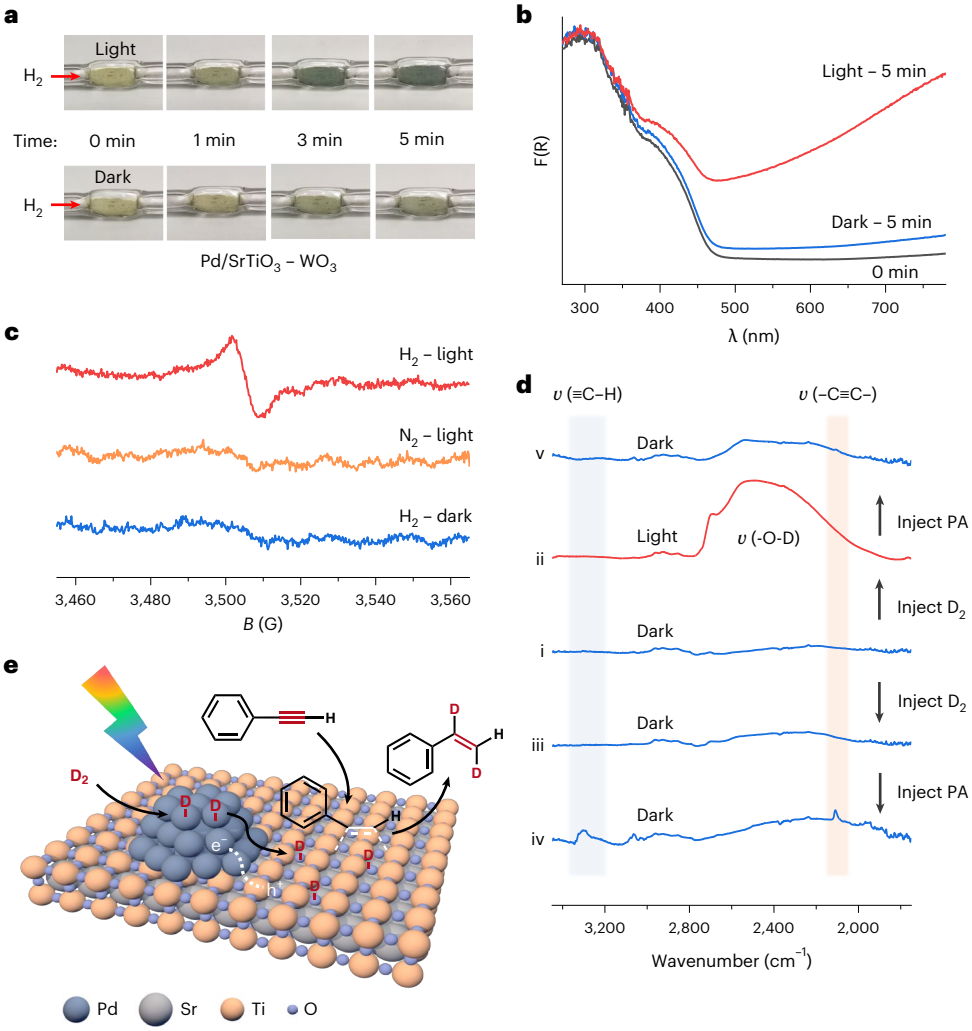

Fig. 5 | Photo-enhanced hydrogen spillover. a,**b**, Colour change (**a**) and corresponding UV–vis DRS spectra (**b**) for a physical mixture of Pd/$SrTiO_3$ and WO_3 before and after treatment with H_2. F is function, R is the diffuse reflectance of the material, F(R) is the transformed function that represents the absorption coefficient. **c**, EPR spectra of Pd/$SrTiO_3$ after treatment with H_2 or N_2. *B* refers to the magnetic field strength applied during the measurement. **d**, In situ FTIR spectra of PA hydrogenation over Pd/$SrTiO_3$. **e**, Schematic diagram of PA hydrogenation over Pd/$SrTiO_3$ under light irradiation.

Mechanism of photo-enhanced spillover hydrogenation

Indeed, the introduction of light irradiation into the reaction system not only efficiently enhances PA adsorption specifically on $SrTiO_3$, but also promotes H_2 activation and facilitates the migration of H* species from Pd to $SrTiO_3$. These improvements probably account for the catalytic semihydrogenation of PA with both good activity and high selectivity under light irradiation (Fig. 2c). To better study this photo-enhanced spillover hydrogenation process, a traditional spillover hydrogenation process was conducted on Pd/$SrTiO_3$–quinoline in the dark. In Fig. 6a and Supplementary Table 1, the activity of Pd/$SrTiO_3$–quinoline in the dark is seen to be rather low due to the inevitable sacrifice of Pd surface for H_2 activation, requiring a reaction time of up to 20 h to achieve complete conversion of PA with a selectivity of 91.5%. In contrast, under light irradiation, complete PA conversion was achieved in just 1 h with a selectivity of 97.6% over Pd/$SrTiO_3$. This finding strongly indicates that light irradiation is advantageous over the surface-poisoning strategy for spillover hydrogenation reactions, as it can ensure complete exposure of the Pd surface for H_2 dissociation and H* formation. Interestingly, this photo-enhanced effect also worked for Pd/$SrTiO_3$–quinoline. Under light irradiation, PA was completely converted within 4 h over Pd/$SrTiO_3$–quinoline with a selectivity of 96.1%, which was much faster than that in the dark condition. This photo-enhanced spillover hydrogenation on Pd/$SrTiO_3$–quinoline was further demonstrated by a evident decrease in E_a (Fig. 6b).

This photo-enhanced spillover hydrogenation was further studied with DFT calculations (Fig. 7a). Under light irradiation, the PA molecule preferentially adsorbed on the surface of the $SrTiO_3$, unlike under dark conditions, where PA strongly adsorbed on the Pd surface. The reaction barriers for the first and second H additions were calculated to be 1.02 and 0.36 eV on the $SrTiO_3$ surface, respectively, compared to 0.84 and 0.37 eV on the Pd surface. This indicates that the photo-induced charged $SrTiO_3$ surface exhibits good catalytic ability for the H-addition reaction. The desorption energy of *$PhCHCH_2$ from $SrTiO_3$ was calculated to be −1.11 eV, indicating that it is a thermodynamically favourable process. This effectively prevents further hydrogenation of *$PhCHCH_2$, resulting in high selectivity towards styrene in PA semihydrogenation. In sharp contrast, *$PhCHCH_2$ was strongly adsorbed on the Pd surface with a desorption energy of 1.12 eV, even higher than the barriers of hydrogenation (0.92 eV). This suggests that the adsorbed *$PhCHCH_2$ on the Pd surface prefers to undergo further hydrogenation to yield ethylbenzene rather than desorption.

Figure 7b shows proposed reaction mechanisms for PA semihydrogenation over Pd/$SrTiO_3$ with or without light irradiation. In the dark, PA is directly adsorbed on the Pd surface and then quickly reacts with the H* produced on the same Pd surface[24]. This makes it difficult

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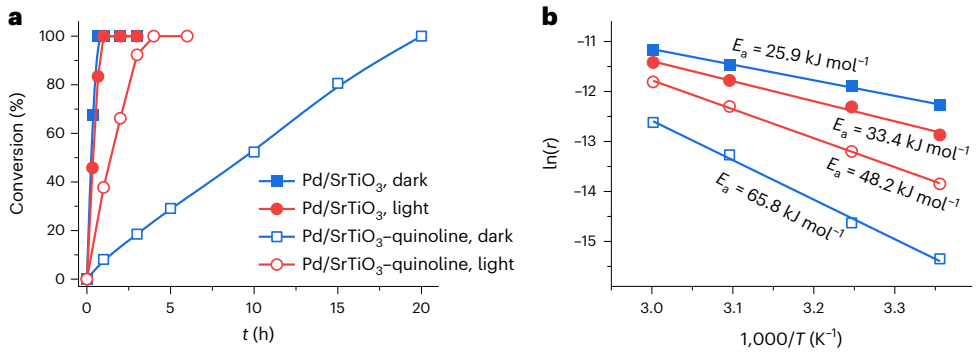

Fig. 6 | Photo-enhanced spillover hydrogenation. a,**b**, PA conversion over Pd/$SrTiO_3$ and Pd/$SrTiO_3$–quinoline with or without light irradiation (**a**) and the corresponding Arrhenius plots (**b**).

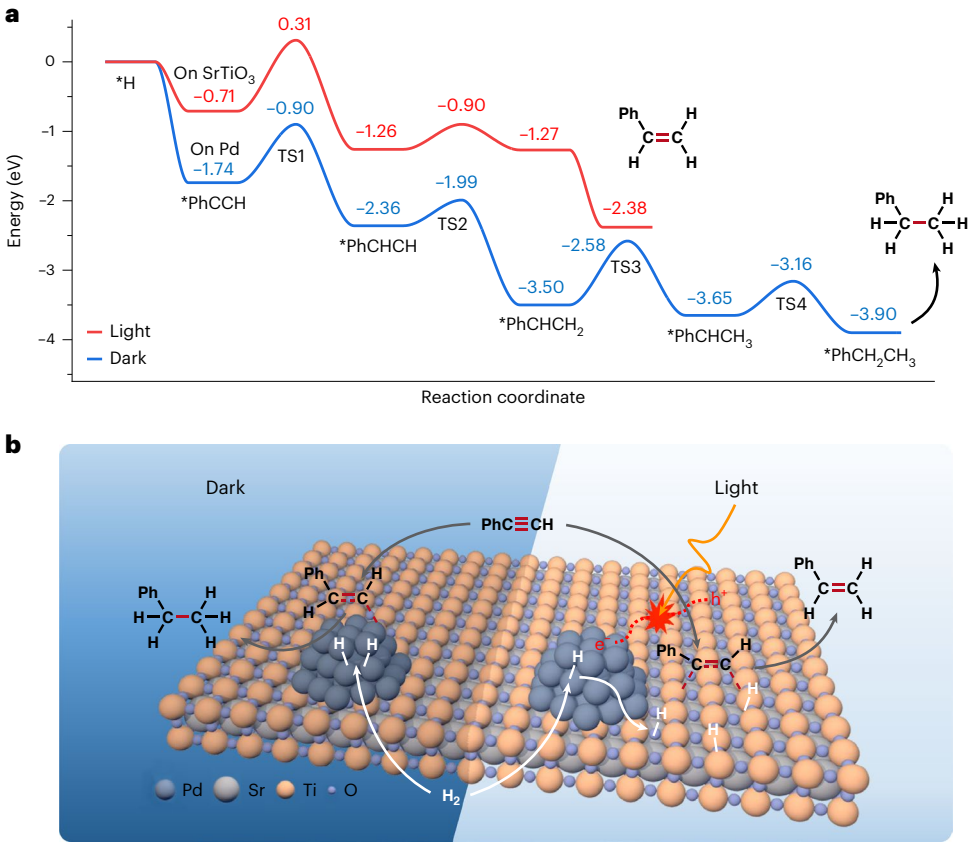

Fig. 7 | Spillover hydrogenation mechanism. a,**b**, Energy profile of PA stepwise hydrogenation (**a**) and a possible reaction mechanism of PA hydrogenation over Pd/$SrTiO_3$ under dark and light conditions (**b**).

for intermediate species to desorb from the Pd surface, resulting in an overhydrogenation issue. Under light irradiation, the photogenerated electrons in the conduction band of $SrTiO_3$ rapidly transfer to bare Pd nanoparticles[45], increasing the electron density on the Pd surface to effectively inhibit PA adsorption and facilitate H_2 dissociation and H* formation, while the photogenerated holes left on $SrTiO_3$ enhance PA adsorption and promote H* migration. The adsorbed PA then reacts with the spilled H* to form styrene, thus achieving a photo-enhanced spillover hydrogenation process over Pd/$SrTiO_3$ under light irradiation.

Universality of photo-enhanced spillover hydrogenation

Various alkynes, including substituted phenylacetylenes and heptyne, were next used as unsaturated substrates for semihydrogenation on Pd/$SrTiO_3$ under light irradiation (Supplementary Fig. 26). As expected, all the substrates were successfully hydrogenated to the corresponding alkenes with high selectivity and good activity. These results clearly suggest that the photo-enhanced spillover hydrogenation process over Pd/$SrTiO_3$ is also effective in the semihydrogenation of a range of alkynes.

Other semiconductor-supported nanocatalysts, such as Pd/ZnO, Pd/PCN and Ru/MnO_x, were prepared for PA semihydrogenation with or without light irradiation (Supplementary Figs. 27–29). The selectivity towards styrene was improved over these supported nanocatalysts under light irradiation. This finding suggests that the photo-enhanced spillover hydrogenation process can also occur on other semiconductor-supported Pd nanocatalysts, facilitating the use

of functional materials to support Pd nanoparticles for the selective hydrogenation of specific target groups, such as C=O, C=N and others.

Conclusions

In summary, we have demonstrated that simply introducing light irradiation to the reaction system can directly induce spillover hydrogenation via site separation during selective hydrogenation reactions over semiconductor-supported Pd nanocatalysts. Under light irradiation, photogenerated electrons accumulating on the bare Pd nanoparticles can greatly suppress the adsorption of alkynes on the nanoparticle surfaces, ensuring that the Pd nanoparticles exclusively facilitate H_2 activation and H* generation, while the photogenerated holes remaining on the support can enhance the adsorption of alkynes on its surface and weaken the H*–support interaction to promote H* migration. This photo-enhanced effect has been experimentally validated using Pd nanoparticles supported on $SrTiO_3$, ZnO and carbon nitride semiconductors, demonstrating good catalytic selectivity and activity in the semihydrogenation of alkynes to alkenes. We hope that the principles of this photo-enhanced effect could be somewhat generalizable for other semiconductor-supported Pd nanocatalysts and could provide a strategy to improve spillover hydrogenation efficiency at large.

Methods

Preparation of Pd/$SrTiO_3$

$SrTiO_3$ was synthesized using a solvothermal method[8], then Pd nanoparticles were immobilized on the $SrTiO_3$ using a photodeposition method. Typically, 0.2 g of $SrTiO_3$ was dispersed in 50 ml of aqueous solution containing methanol (10 vol%) and H_2PdCl_4 solution (10 mg ml^{-1}). After stirring for 30 min under a N_2 atmosphere, the suspensions were irradiated by a xenon lamp for another 30 min. The resulting sample was washed multiple times with deionized H_2O and ethanol, then dried in a vacuum oven at 60 °C for 12 h. The final product was labelled as Pd/$SrTiO_3$. The Pd content was measured to be 0.45 wt% in Pd/$SrTiO_3$ by inductively coupled plasma mass spectrometry (ICP-MS).

Preparation of Pd/$SrTiO_3$–quinoline

For the PA hydrogenation reaction and temperature-programmed desorption measurements, typically, 100 mg of Pd/$SrTiO_3$ was dispersed in 10 ml of acetonitrile containing 50 µl of quinoline. After being stirred at 25 °C for 2 h under a N_2 atmosphere, the resulting mixture was washed multiple times with acetonitrile and then transferred to an oven at 60 °C for 12 h (Pd/$SrTiO_3$–quinoline). For in situ FTIR measurements, Pd/$SrTiO_3$ was sealed in a glass tube and degassed at 120 °C under vacuum for 2 h. When the temperature returned to room temperature, quinoline was injected into the tube in the dark. Once adsorption saturation was reached, the gaseous quinoline was removed (Pd/$SrTiO_3$–quinoline).

Preparation of Pd/$SrTiO_3$ with different Pd particle sizes

Pd single-atom catalyst was synthesized using a solvothermal method[8]. Typically, 10 mmol $C_{16}H_{36}O_4Ti$ was dissolved in 50 ml of $(CH_2OH)_2$ under stirring for 1 h. A 10-ml aqueous solution containing $Sr(NO_3)_2$ at 10 mmol and H_2PdCl_4 at a concentration of 10 mg ml^{-1} was then added dropwise to the solution. After stirring for 1 h, 10 ml of NaOH at a concentration of 50 M was added, and the mixture was stirred for an additional 1 h. The mixture was transferred to an autoclave and heated at 180 °C for 16 h. After cooling to room temperature, the suspension was treated with distilled H_2O and ethanol. The resulting sample was dried at 60 °C overnight, then calcined at 400 °C for 2 h. Pd contents were measured to be 0.49 wt%. Pd/$SrTiO_3$ nanocatalysts with different Pd particle sizes were prepared using a colloidal approach[51]. For the preparation of Pd/$SrTiO_3$ with 2.3 nm Pd, a solution containing $Pd(NO_3)_2$ (0.094 mmol), polyvinylpyrrolidone (PVP, 1.88 mmol, PVP-to-Pd molar ratio of 20:1), $(CH_2OH)_2$ (10 ml) and H_2O (40 ml) was prepared. After refluxing the mixture solution at 60 °C for 3 h, 2.0 g of $SrTiO_3$ was added and stirred for another 2 h. The obtained sample was treated with distilled H_2O and ethanol, then dried in a vacuum oven at 25 °C. Finally, the powder was calcined in flowing air at 550 °C for 3 h with a heating rate of 3 °C min^{-1}. For the preparation of Pd/$SrTiO_3$ with 4.0 and 5.9 nm Pd, the reduction temperature was 80 °C, and the PVP-to-Pd molar ratio was 5 and 1.25, respectively. For the preparation of Pd/$SrTiO_3$ with 10.1 nm Pd, PVP was not utilized, and the refluxing temperature was set at 90 °C.

Preparation of Pd/ZnO

ZnO was also synthesized using a solvothermal method[52], and Pd/ZnO (0.43 wt% Pd) was prepared by the same process as used for Pd/$SrTiO_3$ preparation.

Preparation of Pd/PCN

Polymeric carbon nitride (PCN) was synthesized by directly annealing 10 g of melamine at 550 °C in air for 2 h. Pd/PCN (0.47 wt% Pd) was prepared by the same process as used for Pd/$SrTiO_3$ preparation.

Characterization

X-ray diffraction, UV–vis DRS, XPS and EPR measurements were performed on a Bruker D8 Advance diffractometer, a Varian Cary 500 scanner, a Thermo ESCALAB250 spectrometer and a Bruker model A300 spectrometer, respectively. Analysis of metal content was carried out with a PerkinElmer Avio 200 ICP optical emission spectrometer. Aberration-corrected STEM measurements were performed with a Thermo Fisher Themis Z instrument. Fluorescence mode was used to collect X-ray absorption spectroscopy data for the Pd K-edge in a home-made in situ photocatalytic cell using a silicon drift detector at the BL44A beamline of TPS, National Synchrotron Radiation Research Center (NSRRC), Taiwan.

Computational details

The Vienna Ab initio Simulation Package (VASP)[53] with the projector augmented wave (PAW) method was used for DFT calculations[54]. Generalized gradient approximation (GGA) with the Perdew–Burke–Emzerhof (PBE) functional was used to treat the exchange functional[55]. A value of 450 eV was set for the energy cutoff in the planewave basis expansion. Partial occupancies of the Kohn–Sham orbitals were allowed using the Gaussian smearing method and a width of 0.2 eV. The Brillouin zone sampled with the Monkhorst–Pack k-point of 1 × 2 × 1 and 2 × 2 × 1 was used for $SrTiO_3$ and Pd, where the spin-polarized effect was considered. Surface (110) with the $p(2 \times 1)$ unit cell of $SrTiO_3$ was used[56]. A four-layer slab with a $p(2 \times 2)$ supercell was used to represent the Pd (111) surface. The self-consistent calculations applied a convergence energy threshold of 10^{-5} eV, and the force convergency was set to 0.05 eV $Å^{-1}$. The vacuum layer was set to be 15 Å above the surface. To simulate the effect of photogenerated electrons on the adsorption of PA, different amounts of electrons (2, 4, 6 and 8) were added to the Pd(111) surface. To simulate the effect of photogenerated holes on $SrTiO_3$, some oxygen atoms were removed from the surface of $SrTiO_3$ to create oxygen vacancies, as the spilled-over H* always induce oxygen vacancies on the support. The adsorption energies of C_8H_6 (E_{ad}) were estimated by equation (1):

$$E_{ad} = E_{sur+gas} - E_{sur} - E_{gas} \quad (1)$$

Here, E_{sur} represents the energy of the clean surface, E_{gas} denotes the energy of a free PA and styrene molecule in the vacuum, and $E_{sur+gas}$ is the energy of a gas molecule adsorbed on the surface.

The free energy corrections were considered at 298 K, following equation (2):

$$\Delta G = \Delta E + \Delta GZPE + \Delta GU - T\Delta S \quad (2)$$

where ΔE, ΔGZPE, ΔGU and ΔS refer to the DFT-calculated energy change, the correction from the zero-point energy, the correction from the inner energy and the correction from entropy[57].

Catalytic hydrogenation reaction under dark conditions

The catalytic semihydrogenation of PA was performed in a 50-ml quartz batch reactor, using a water bath equipped with circulating water to maintain the reaction temperature at 25 °C. First, 20 mg of catalyst was dispersed in 5 ml of acetonitrile while stirring, with N_2 bubbling for 30 min. Second, 100 µmol of PA was added, and the hydrogenation reaction was carried out under a 1-bar atmosphere of 20% H_2/Ar.

Catalytic hydrogenation reaction under light conditions

The procedure was identical to that under dark conditions, except that the reaction solution was irradiated with a light-emitting diode (LED) light source throughout the hydrogenation process. The wavelengths and power densities of the LED lights were tested by a National Institute of Standards and Technology (NIST) traceable light measurement system (International Light Technologies). A water bath equipped with circulating water was used to minimize the potential heat effect. A gas chromatograph (Shimadzu GC-2030) equipped with a flame ionization detector (FID) was used for analysis of the products. The catalytic performance of PA hydrogenation, including conversion and selectivity towards styrene, was calculated based on equations (3) and (4), respectively:

$$\text{Conversion}\,(\%) = [(C_0 - C_P)/C_0] \times 100\,\% \quad (3)$$

$$\text{Selectivity}\,(\%) = [C_S/(C_0 - C_P)] \times 100\,\% \quad (4)$$

where C_0, C_P and C_S are the initial concentration of PA, the remaining concentration of PA and the concentration of produced styrene, respectively.

Data availability

Data that support the findings of this study are available within the Article and its Supplementary Information, or available from the corresponding authors upon request. Source data are provided with this paper.

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Author contributions

X.W. and J.Z. conceived and supervised the research project. J.Z., K.L. and B.Y. designed and performed the synthesis and catalytic experiments. J.-J.M., Y.-Y.C. and S.-F.H. carried out the X-ray absorption spectroscopy experiments. S.Z. was involved in part of the material characterizations. X.W., J.Z., J.C.Y., M.A., Y.H., M.H., K.L. and B.Y. prepared the manuscript. All authors contributed to the overall scientific interpretation.

Competing interests

The authors declare no competing interests.

Additional information

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